

MUHAMMAD AALIYAN ARIF

AI / ML Engineer

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PROFESSIONAL SUMMARY

Targeting a full-time AI/ML Engineer role. Builds end-to-end machine learning systems — from data pipelines and model training to explainability and production APIs — bringing applied Machine Learning, Deep Learning, Computer Vision, and NLP together with full-stack engineering to ship clean, tested, production-ready products.

EXPERIENCE

Junior Full Stack Developer — Big Binary Tech

Apr 2026 – May 2026

- Delivered three production-ready features across React frontend and Django backend as a full team member, consistently meeting all sprint deadlines.
- Reduced bug-fix turnaround by independently diagnosing and patching both React components and Django API endpoints; documented workflows that cut onboarding setup time ~30%.

Frontend Development Intern — Code Alpha

Jul 2025 – Oct 2025

- Built and deployed four responsive React + Tailwind web apps, each passing client acceptance criteria on first review.
- Extracted 12+ shared UI components (~40% less duplicate code) and improved core-page UX, cutting UI-related support tickets by 25%.

Teaching Assistant, Machine Learning — UMT

Apr 2025 – Jul 2025

- Mentored 30+ students through Python model-training pipelines (TensorFlow, Scikit-learn), driving a 0.4 GPA-point average grade improvement.
- Resolved 90% of student debugging issues live during labs, reducing after-hours help requests to near zero.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript (ES2022+), SQL
AI / ML & DL	TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy
Vision & NLP	YOLOv8, OpenCV, NLTK, spaCy, LangChain, RAG pipelines, LLM APIs, AIML, Prolog (pyswip)
Frontend	React, Next.js, Tailwind CSS, shadcn/ui, TanStack Query, Zustand, React Router
Backend & APIs	FastAPI, Django, Django REST Framework, Flask, Express.js, REST API Design, JWT Auth
Data & Tools	Neo4j (Cypher), SQLite, Plotly, Matplotlib, Seaborn, Git, GitHub, Vercel, Pytest, Postman

PROJECTS

Worker Safety Monitor — Real-Time PPE & Danger-Zone Compliance System

2026

YOLOv8, ByteTrack, OpenCV, FastAPI, WebSockets, SQLite, React, TypeScript, Three.js

- Built an end-to-end computer-vision safety system detecting PPE compliance and danger-zone dwell violations from a live camera feed, combining a custom-trained YOLOv8 detector with ByteTrack tracking and temporal smoothing to suppress false positives.
- Shipped a FastAPI + WebSocket backend streaming live frames and violations to a React/Three.js control room with live overlays, analytics dashboards, and 3D replay, backed by 315 automated tests.

VERIDEX — AI Research & Evidence Intelligence Platform

2026

Python, FastAPI, Next.js, PostgreSQL, Qdrant, Redis, Celery, Ollama, sentence-transformers

- Built a local-first, agentic RAG platform for evidence-grounded literature research, orchestrating planner, researcher, evidence, and citation agents over a hybrid vector + BM25 retrieval pipeline with cross-encoder reranking.
- Engineered an anti-fabrication layer that verifies every AI-proposed quote against its source before acceptance, computing deterministic SUPPORTED/CONTRADICTED verdicts from evidence weights rather than model opinion.

VisuaPrompt — Automated ML & Data Analytics Platform

2025

FastAPI, React 19, scikit-learn, XGBoost, LightGBM, spaCy, Plotly, SQLAlchemy, JWT

- Built an end-to-end, no-code data science platform carrying a raw dataset from upload through automated cleaning, feature engineering, AutoML training, prediction, and explainability — served by a single JWT-secured FastAPI backend of 37 REST endpoints and a 13-page React 19 UI, validated end-to-end with Pytest.
- Engineered an AutoML pipeline that auto-detects task type and benchmarks 13+ algorithms (scikit-learn, XGBoost, LightGBM), plus a spaCy/NLTK natural-language query engine for plain-English dataset queries and interactive visualizations.

Knowledge-Based Conversational AI System

2024

Python, Flask, Neo4j, AIML, Prolog, NLTK

- Engineered a hybrid AI chatbot unifying three reasoning paradigms — AIML rule-based dialogue, Prolog logic inference via pyswip, and Neo4j graph memory — enabling multi-hop relationship queries beyond flat SQL schemas.
- Implemented NLTK POS tagging and parsing to extract intent and route queries to the right engine, and Cypher templates that cut average graph traversal time by 35% versus naive traversal.

EDUCATION

BS, Artificial Intelligence — University of Management & Technology (UMT) 2021 – 2025

CGPA: 3.5 / 4.0 | Specialization in ML & Intelligent Systems